Explainable systems

Ethics & bias in AI

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Fri[illegible]nens Fritz / Better Images of AI / GPU shot etched 5 / CC-BY 4.0

Why Explainability for AI?

- Regulatory Requirements
  - E.g. GDPR's "right to explanation”, EU AI Act
- Detecting and mitigating bias in algorithms
- Ensuring fairness across different demographic groups
- Supporting human autonomy by enabling informed consent
- Providing recourse for affected individuals
- User trust & confidence
- Better model development

Transparency in the EU AI Act

“transparency means that AI systems are developed and used in a way that **allows appropriate traceability and explainability**, while **making humans aware** that they communicate or interact with an AI system, as well as **duly informing deployers** of the **capabilities** and **limitations** of that AI system **and affected persons** about their **rights**” (Recital 27).

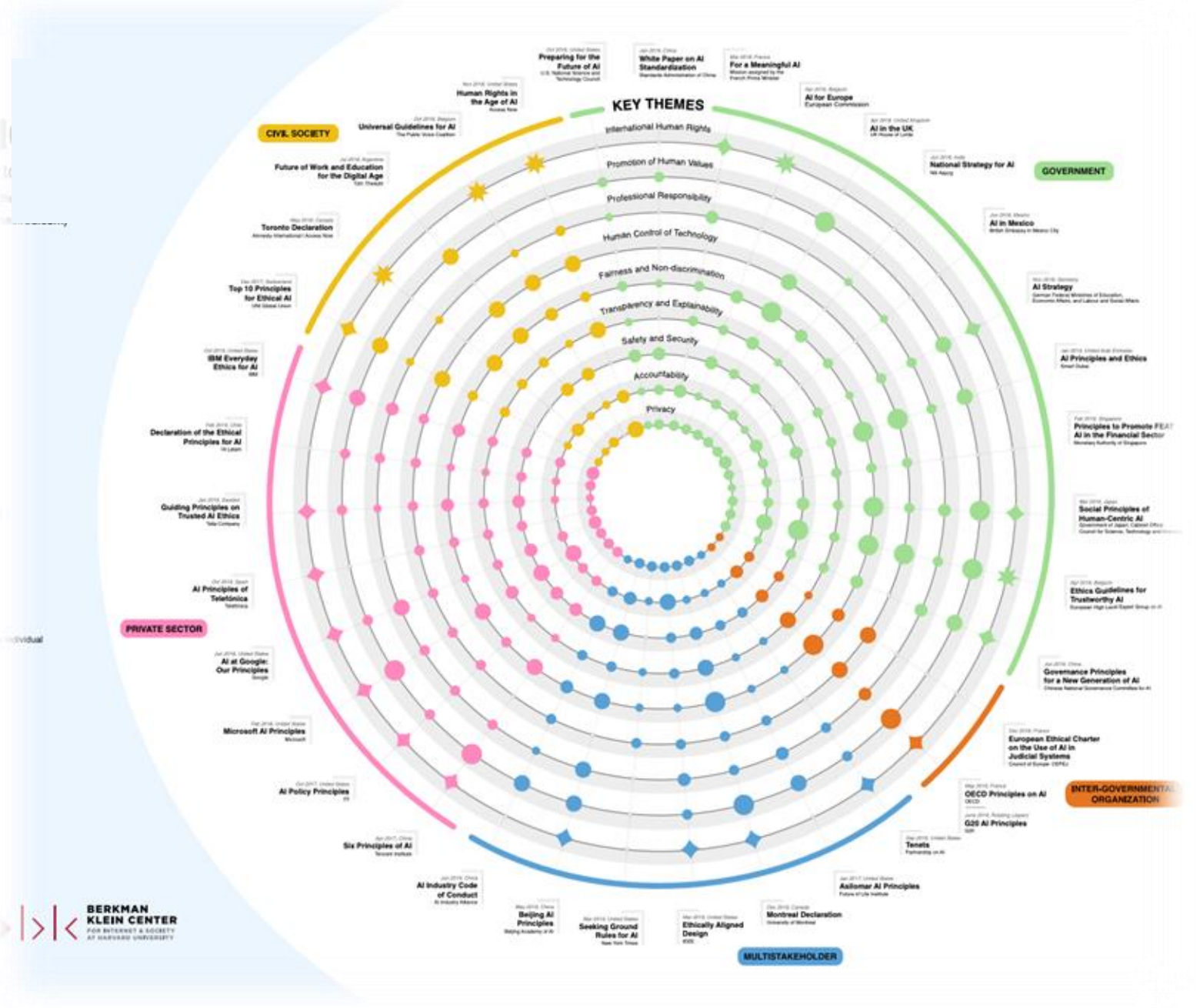
Source: https://www.euaiact.com/key-issue/5

Principles for Responsible AI

Berkman Klein Centre (2020)

Key themes:

1. Privacy
2. Accountability
3. Safety and security
4. Transparency and explainability
5. Fairness and non-discrimination
6. Human control of technology
7. Professional responsibility
8. Promotion of human values
9. International Human Rights



Fjeld, J., Achten, N., Hilligoss, H., Nagy, A., & Srikumar, M. (2020). Principled artificial intelligence: Mapping consensus in ethical and rights-based approaches to principles for AI. *Berkman Klein Center Research Publication*, (2020-1). https://cyber.harvard.edu/publication/2020/principled-ai

UK Gov A pro-innovation approach to AI regulation

- Safety, security & robustness
- Appropriate transparency and explainability
- Fairness
- Accountability and governance
- Constestability and redress

UNESCO Recommendations on the ethics of AI (2021)

- Proportionality and Do No Harm
- Safety and security
- Fairness and non-discrimination
- Sustainability
- Right to Privacy, and Data Protection
- Human oversight and determination
- Transparency and explainability
- Responsibility and accountability
- Awareness and literacy
- Multi-stakeholder and adaptive governance and collaboration

Interpretability vs. Explainability

Varying definitions in the literature, here we will use:

Interpretability refers to the characteristic of a model that allows humans to understand its inner workings and reasoning process.

An interpretable model is one where:

- The decision-making process is transparent by design
- Humans can trace the path from input to output
- The model's parameters have intuitive meanings

Examples: Decision trees, linear regression, rule-based systems

Interpretability vs. Explainability

Explainability refers to the active process of explaining a model's behaviour, often through supplementary methods or techniques.

Explainability:

- Can be applied to "black box" models after they're built
- Provides justifications or rationales for specific decisions
- Doesn't necessarily reveal the full mechanism of the model

Examples: Feature importance scores, surrogate models

Why explainability matters

- **Complex AI models (e.g., deep learning, ensemble methods)** are often difficult to interpret.
- Bias can be present in training data, feature importance, or decision-making logic.

Explainability helps debug fairness interventions

- Helps **identify and mitigate** biases in model predictions.
- Aids regulators and policymakers in **auditing AI fairness**.

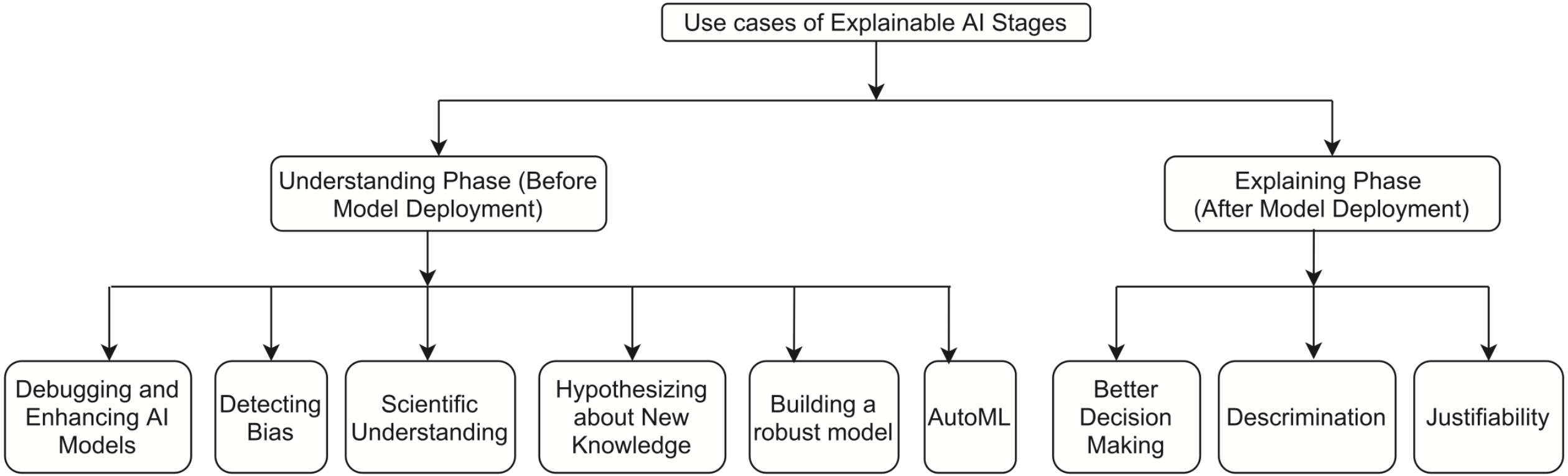
What goes into an explanation?

Information Commissioners Office (2022) identified six main types of explanation:

- **Rationale explanation:** the reasons that led to a decision, delivered in an accessible and non-technical way.
- **Responsibility explanation:** who is involved in the development, management and implementation of an AI system, and who to contact for a human review of a decision.
- **Data explanation:** what data has been used in a particular decision and how. Fairness explanation: steps taken across the design and implementation of an AI system to ensure that the decisions it supports are generally unbiased and fair, and whether or not an individual has been treated equitably.
- **Safety and performance explanation: steps** taken across the design and implementation of an AI system to maximise the accuracy, reliability, security and robustness of its decisions and behaviours.
- **Impact explanation:** steps taken across the design and implementation of an AI system to consider and monitor the impacts that the use of an AI system and its decisions has or may have on an individual, and on wider society.

Source: https://ico.org.uk/media/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/explaining-decisions-made-with-artificial-intelligence-1-0.pdf

Use cases of XAI (Dwivedi et al, 2023)



Dwivedi, R., Dave, D., Naik, H., Singhal, S., Omer, R., Patel, P., ... & Ranjan, R. (2023). Explainable AI (XAI): Core ideas, techniques, and solutions. *ACM Computing Surveys*, *55*(9), 1-33.

Types of Explanations: Global vs. Local

Global explanations provide understanding of the model's overall behaviour:

- Feature importance rankings across the entire dataset
- Rule extraction describing general patterns
- Variable interaction effects
- Use case: Auditing models for regulatory compliance

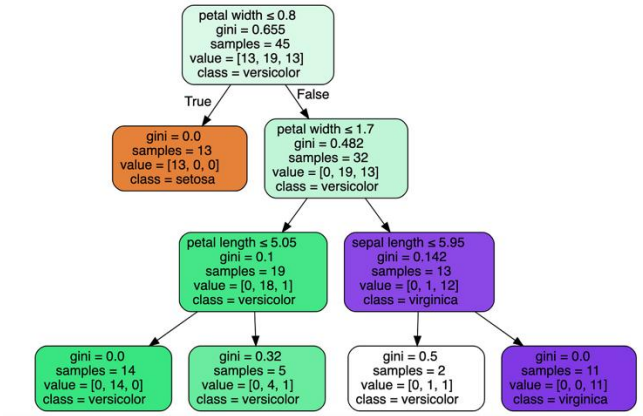
Local explanations focus on individual predictions:

- Why a specific input received its particular output
- What features were most influential for a single decision
- Counterfactual explanations for specific instances
- Use case: Providing feedback to affected individuals

Types of Explanations: Ante-hoc vs. Post-hoc

Ante-hoc methods (also called intrinsic or transparent methods):

- Build transparency into the model from the beginning
- Choose inherently interpretable model architectures
- May trade some performance for interpretability
- Examples: Decision trees, sparse linear models, rule lists



Post-hoc methods:

- Apply explanatory techniques after model training
- Can work with any model, including black-box systems
- Don't modify the underlying model
- Examples: LIME, SHAP, partial dependence plots, attention visualisation

Feature relevance attribution methods

(a) Heatmap [168] (b) Attribution [293] (c) Grad-CAM [292]

Source Arietta et al, 2020: https://doi.org/10.1016/j.inffus.2019.12.012

Techniques for Explainability

SHAP & LIME – Feature Attribution methods:

- **SHAP (Shapley Additive Explanations)**
  - Measures how much each feature **contributes** to a prediction.
  - Useful for **detecting biased features** influencing outcomes.
  - **Example:** In a hiring model, if “Gender” has a high SHAP value, it means the model is unfairly influenced by it.
- **LIME (Local Interpretable Model-agnostic Explanations)**
  - Creates **local surrogate models** to explain individual predictions.
  - Helps in detecting **case-by-case bias**.
  - **Example:** Why was a specific female applicant rejected for a job? LIME can reveal if “Years of Experience” was weighed unfairly for women.

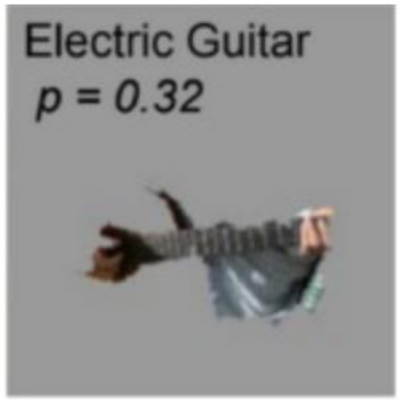
LIME (Local Interpretable Model-agnostic Explanations)

How LIME works:
For a specific prediction instance, LIME generates perturbed samples around this instance (by randomly changing feature values)

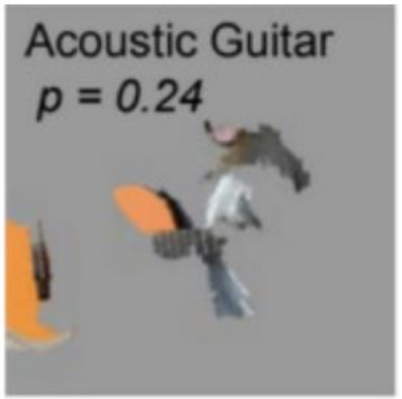
- The original black-box model makes predictions on these perturbed samples
- LIME assigns weights to these samples based on their proximity to the original instance
- A simple model (typically linear regression) is trained on these weighted samples
- The coefficients of this simple model provide insights into which features are most important for the specific prediction

LIME

(a) Original image



(b) Explaining *electric guitar*

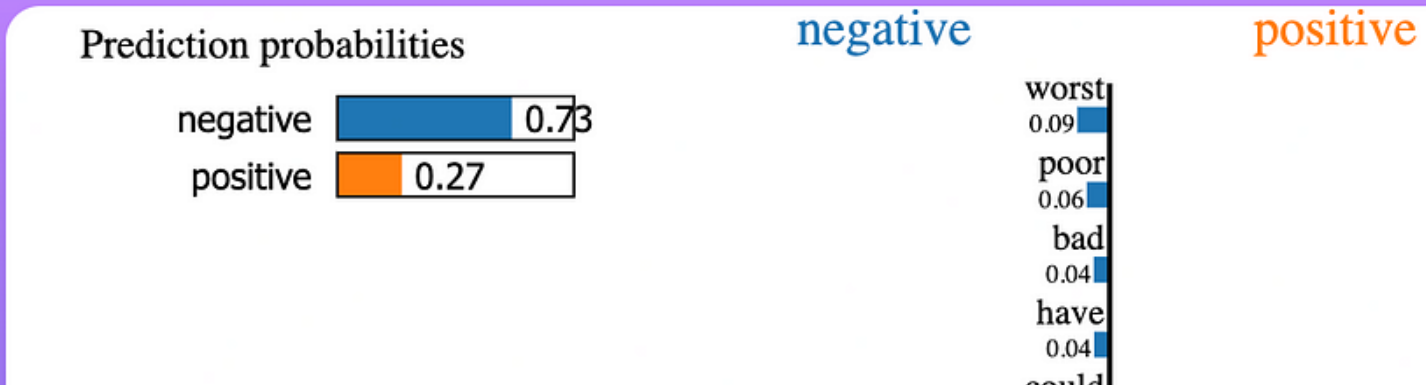


(c) Explaining *acoustic guitar*

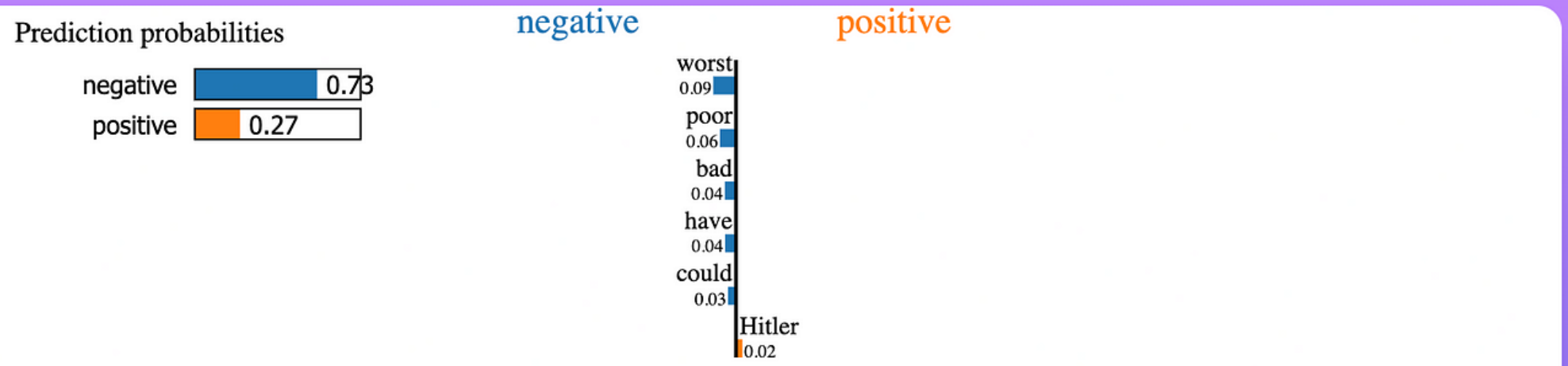
Can be adapted for different data types (text, images, tabular data)

Limitations:

- Explanations can be unstable (different runs might give different explanations)
- Selection of neighbourhood size affects explanation quality
- Computationally expensive for complex instances



Example of LIME outputs



Text with highlighted words

This could well be the worst film I've ever seen. Despite what Mikshelt claims, this movie isn't even close to being historically accurate. It starts badly and then it's all downhill from there. We have Hitler's father cursing his own bad luck on the "fact" that he'd married his niece! They were in fact, second cousins. Hitler's mother, Klara, called his father, Alois, "uncle" because Alois had been adopted and raised by Klara's grandfather and brought up as his son, when he was really his nephew. Alois was much older than Klara and so as a child she'd got into the habit of calling Alois, "uncle."|br /||br /|The scene in the trenches where Hitler is mocked by his fellow soldiers and decides to take it out on his dog is simply a disgrace and an insult to the intelligence of all viewers. We see Hitler chase the dog through the trench, when he catches up with the poor thing he proceeds to thrash it for disobeying him. In the distance we see and hear his fellow soldiers continue to mock and chastise the cowardly little man, but then a shell lands directly on his persecutors, and every last one, we are told, is killed outright. How then, if Hitler was the only person to survive the scene, did this tale of brutality and cowardice come to be told? Did Hitler himself go around "boasting" about it? - I don't think so.|br /||br /|Next up, Hitler bullies and intimidates a poor, stressed out and war weary Jewish officer into giving him an Iron Cross! I can only assume that this Jewish officer had been a pawnbroker before fighting for the Fatherland, and had thoughtfully brought along some pledged medals from his shop,

SHAP (SHapley Additive exPlanations)

How SHAP works:

SHAP combines game theory with local explanations to allocate feature importance values that add up to the difference between the actual prediction and the average prediction.

- Based on cooperative game theory's Shapley values
- Considers all possible feature combinations (coalitions)
- Computes how much each feature contributes to the prediction when added to different feature subsets
- Assigns importance values that sum to the difference between the model output and the average output

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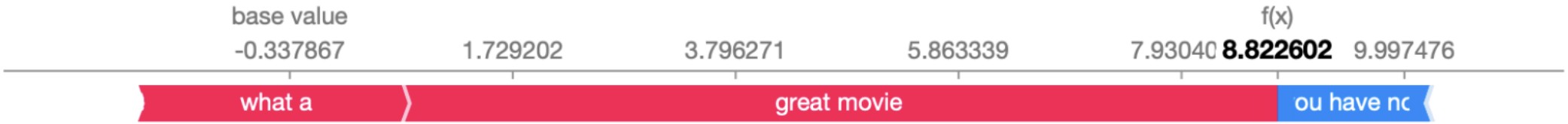
SHAP



what a great movie ! . . . if you have no taste .

SHAP (SHapley Additive exPlanations)

What challenges might arise from computing all feature subsets in large datasets?

Limitations:

- Computationally intensive
- Requires careful selection of the baseline/reference value
- Assumes feature independence for computational tractability

Counterfactual explanations

Counterfactual Explanations – "What If" Analysis for Bias Impact

- **What if we change a sensitive attribute?** Would the prediction change?
- **Example:** If a **loan application** is rejected, would changing the applicant's gender from **female to male** result in approval?
- Helps identify cases where **sensitive attributes disproportionately impact outcomes**.

→ Google what if tool support this – Explore here

Post-model explanations don’t always make sense in a human-understandable manner

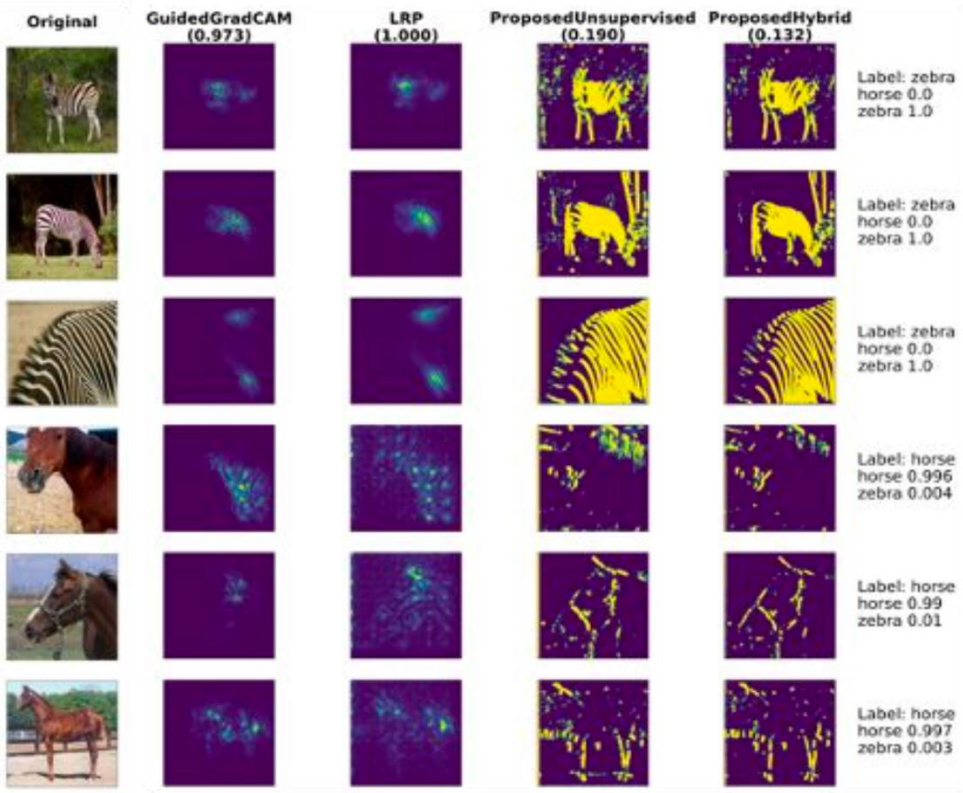
Post-model methods presented for computer vision
- most produce some kind of saliency-maps

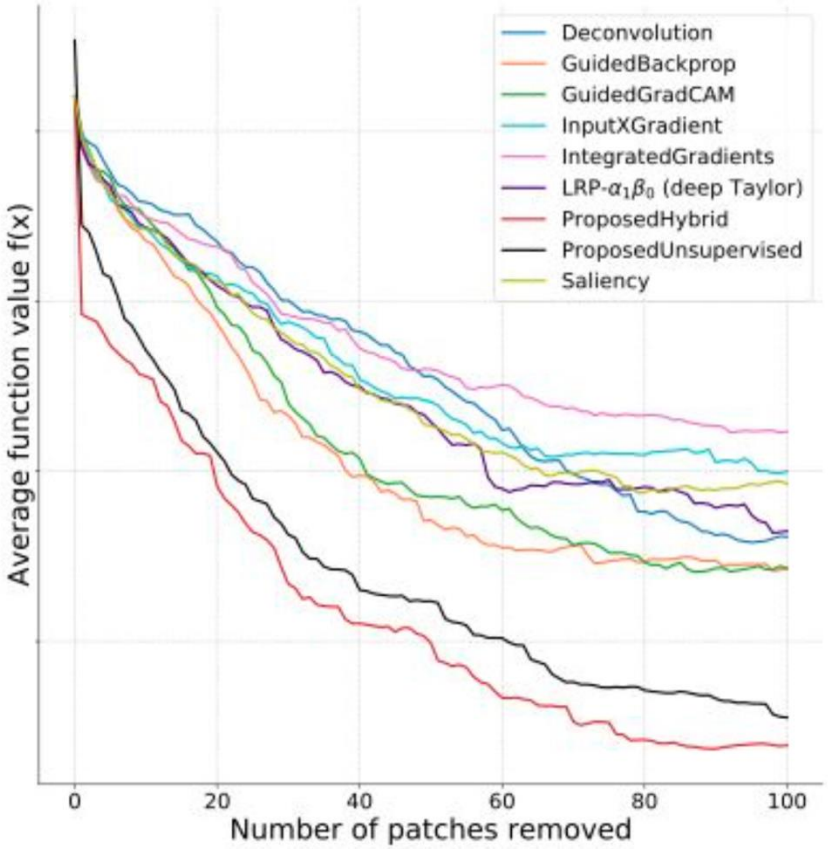
Are the outputs considered meaningful explanations for humans?

Figure - Do post-model explanations make sense? (Image from [1])

Sources: [1] [Cynthia Rudin “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead”]

In-model explanations: Jointly learn to classify and explain





- Rio-Torto et al https://www.sciencedirect.com/science/article/abs/pii/S0167865520301240

Can we show that post-model methods generate misleading explanations?

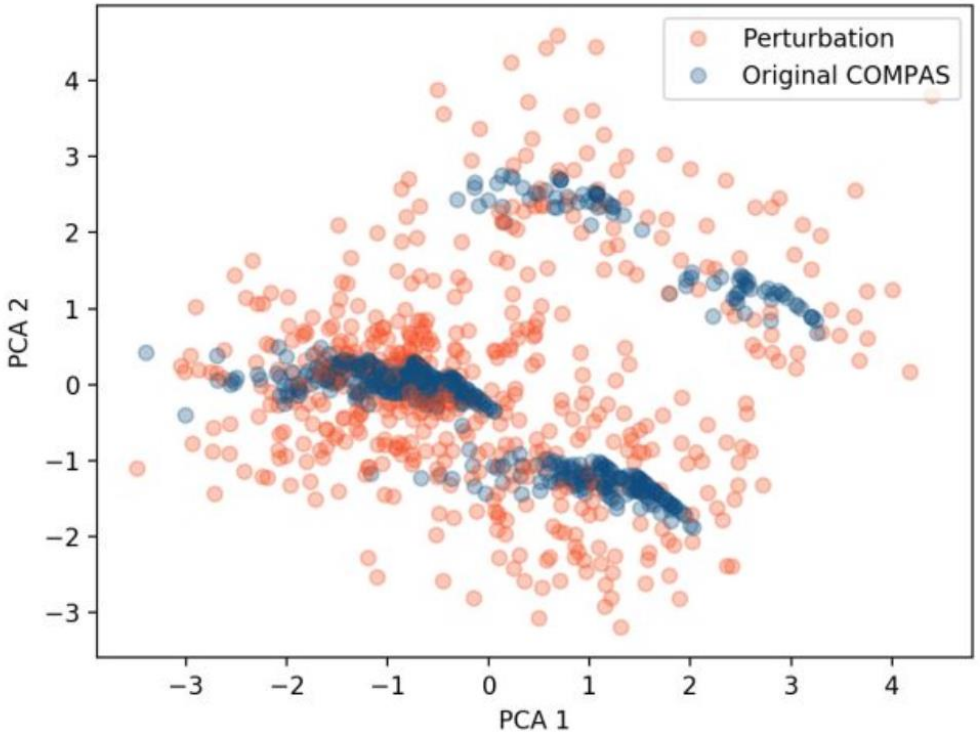
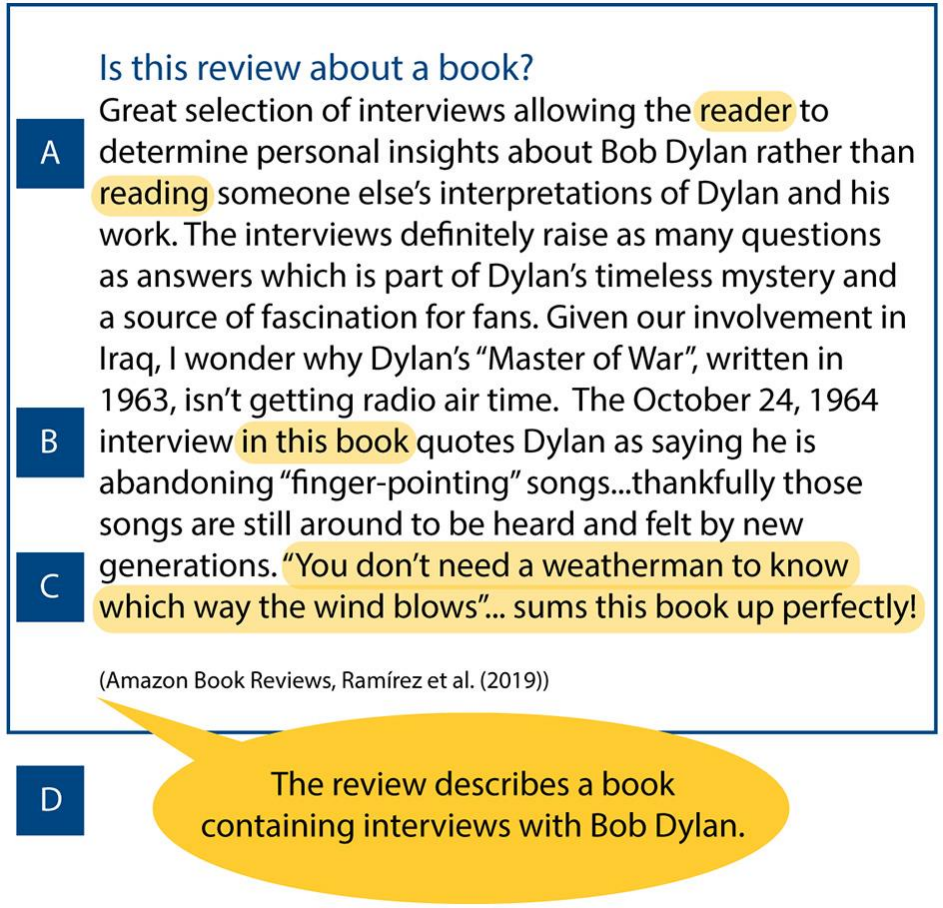


Figure - Fooling LIME and SHAP (Image from [1])

Slack et al, https://arxiv.org/abs/1911.02508

Annotating for explanations



https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2024.1260952/full

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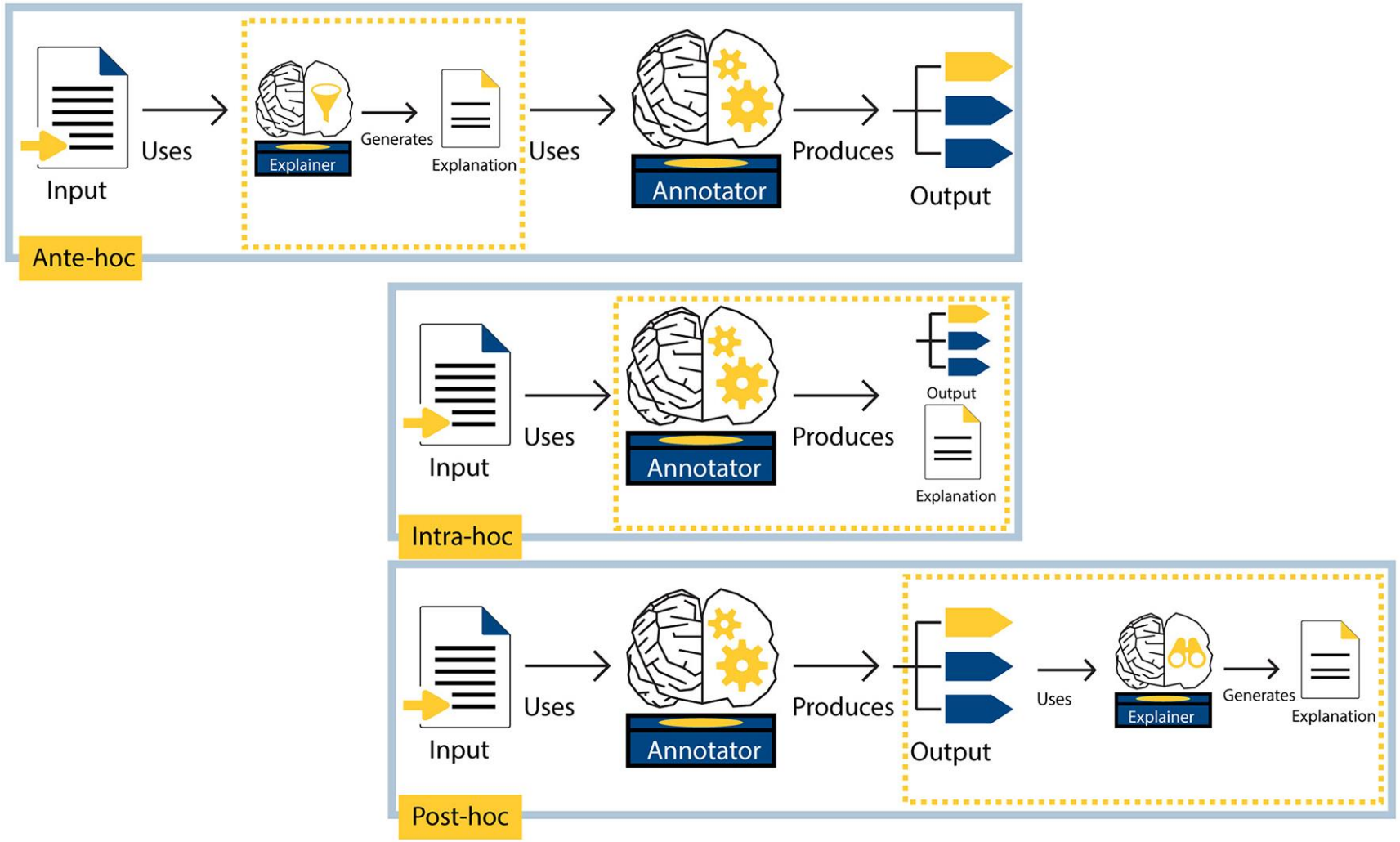

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What makes a good explanation?

How can we evaluate explanations?

Evaluation Criteria

- **Fidelity** – Does the explanation accurately reflect the model's true reasoning?
- **Interpretability** – Is the explanation easy for humans to understand?
- **Consistency** – Do similar inputs yield similar explanations?
- **Completeness** – Does the explanation provide enough information for understanding?
- **Actionability** – Can users make informed decisions based on the explanation?

How can we evaluate explanations?

Evaluation Methods

- **Human-Centred Evaluations**
  - User studies – how well do explanations help users understand and trust the AI?
  - Cognitive load analysis –how much effort is needed to process the explanation?
- **Model-Based Evaluations**
  - Fidelity testing – Compare explanations with actual model behaviour (e.g., feature importance vs. impact on predictions).
  - Perturbation analysis – Modify inputs slightly and observe changes in explanations.
- **Quantitative Metrics**
  - Sufficiency & comprehensiveness – How much of the explanation is necessary to reproduce the decision?
  - Sparseness & simplicity – Favouring concise, less complex explanations (e.g., LIME, SHAP scores).

Explanations for who?

Model Developers

Domain Experts

Regulators and Auditors

End Users

Affected Individuals

Explanations for who?

Regulators and Auditors:

- Need: Evidence of compliance with standards and fairness requirements
- Format: Standardized documentation and formalized explanations
- Preferred tools: Model cards, datasheets, fairness metrics, impact assessments
- Level of detail: Comprehensive with focus on potential harms and safeguards

Explanations for who?

Model Developers:

- Need: Detailed technical explanations to debug and improve models
- Format: Feature interactions, error analysis, edge case identification
- Preferred tools: Technical visualizations, performance metrics, sensitivity analysis
- Level of detail: High technical complexity is acceptable

Domain Experts:

- Need: Validation that the model aligns with domain knowledge
- Format: Explanations in domain-specific terminology
- Preferred tools: Feature importance aligned with domain concepts, case-based reasoning
- Level of detail: Moderate technical complexity with domain relevance

Explanations for who?

End Users:

- Need: Actionable understanding that builds appropriate trust
- Format: Simple, non-technical explanations focused on key factors
- Preferred tools: Visualizations, counterfactual explanations, confidence indicators
- Level of detail: Low technical complexity, high relevance to user goals

Affected Individuals:

- Need: Understanding why a decision was made about them specifically
- Format: Personalized explanations with actionable feedback
- Preferred tools: Counterfactual explanations ("what would need to change")
- Level of detail: Focused on personal relevance and recourse

Summary

- Different stakeholders will have different requirements and levels of knowledge which should be acknowledge in explanation design
- Existing methods help to explain different aspects and a range of approaches may be needed
- Inherently explainable models may be more reliable for high-risk use contexts
- Novel methods for generating explanations about how models and systems function are needed to improve transparency